Explainable Graph Neural Networks & Hybrid Graph Databases for Brain Biomarker Discovery

Part 2: Pipeline Execution and Results

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M.Tech. / Ph.D. Course Project Presentation — November 2026



Recap, Problem, Idea and Initial Features

A. Recap: Problem Statement & Core Concept

Functional disconnection in resting-state fMRI appears years before structural atrophy; goal is early detection from connectome alone.

Task: Graph-level binary classification into Healthy (0) vs Neurodegenerative (1):

Plus extraction of a post-hoc biological explanation subgraph $G_s \subseteq G_k$.

Each subject brain is a graph $G_k=(V,E_k,X_k)$ on the AAL-116 atlas with $|V|=116$ anatomical regions and thresholded correlation edges.

Core architectural pipeline from Deck 1: Edge-weight-aware GCN $\to$ GATv2 dynamic attention with continuous correlation attributes $\to$ Dual pooling [mean, max] $\to$ SwiGLU classification head $\to$ GNNExplainer.

B. Initial Features Selected & Engineering

Topological Descriptors Computed (5 node + 1 edge):

Level	Feature	How it is computed
Node (5-dim)	Degree	k_i / max_j k_j (binary neighbor ratio)
Node	PageRank	Power iteration on |z_ij|, damping 0.85
Node	Clustering	Weighted geometric-mean triangle intensity
Node	Betweenness	Normalised shortest-path betweenness
Node	Eigenvector	Leading eigenvector of |W|, max-scaled
Edge (1-dim)	z_ij	Signed z-score (GATv2); |z_ij| for GCN

Initial Engineering Flow:

Global-signal removal $\to$ z-scoring off-diagonal entries $\to$ pruning $|z_{ij}| < 0.3$ $\to$ symmetric COO edge list $\to$ NetworkX / Neo4j-GDS centralities.

Critical Threshold Clarification: In Deck 1, threshold was stated as $|r_{ij}| \ge 0.3$. In our actual code, it is applied **after** z-scoring: $|z_{ij}| \ge 0.3$. This keeps ~76% density (5,092 edges per brain).

Pipeline Step 1: Data Collection & Cohort Generation

A. Cohort Selection Rationale

Clinical cohorts (ADNI, PPMI) require signed institutional data-use agreements that remain pending approval.

To build, debug, and validate the full deep geometric pipeline end-to-end, we generated a **synthetic AAL-116 cohort with a planted, mathematically verifiable disease signature**.

Cohort Scale: $N=500$ subjects (250 Healthy, 250 Diseased, strictly alternating by index), ages 50–84.

Multi-Scanner Variation: Assigned round-robin across 3 imaging sites (Site A, Site B, Site C).

The Planted Ground-Truth Advantage: Because the disease signature is explicitly engineered into the connectome, we can mathematically evaluate **explanation faithfulness** against ground truth. Real patient data cannot provide this objective baseline.

B. Generative Model (`generate_synthetic.py`)

Base Functional Covariance & Correlation:

Planted Disease Perturbation (Diseased subjects only):

Scanner Site Noise Injection:

Scanner Site	Noise Level (σ_s)	Subject Count
Site A	0.02 (Low noise)	167 subjects
Site B	0.04 (Moderate)	167 subjects
Site C	0.06 (High noise)	166 subjects

Pipeline Step 2: Cleaning, Preprocessing & Leakage Control

A. Four Transforms per Subject (`transform.py`)

1. **Global Signal Removal**: Subtract mean off-diagonal correlation, diagonal reset to 1:

2. **Per-Subject Z-Scoring**: Zero-mean, unit-variance on off-diagonals; diagonal set to 0:

3. **Sparsification & Edge Export**: Upper triangle $i<j, W_{ij}\ne 0 \to$ `edge_list.csv` (2,545,839 rows) loaded into Neo4j via `LOAD CSV`.

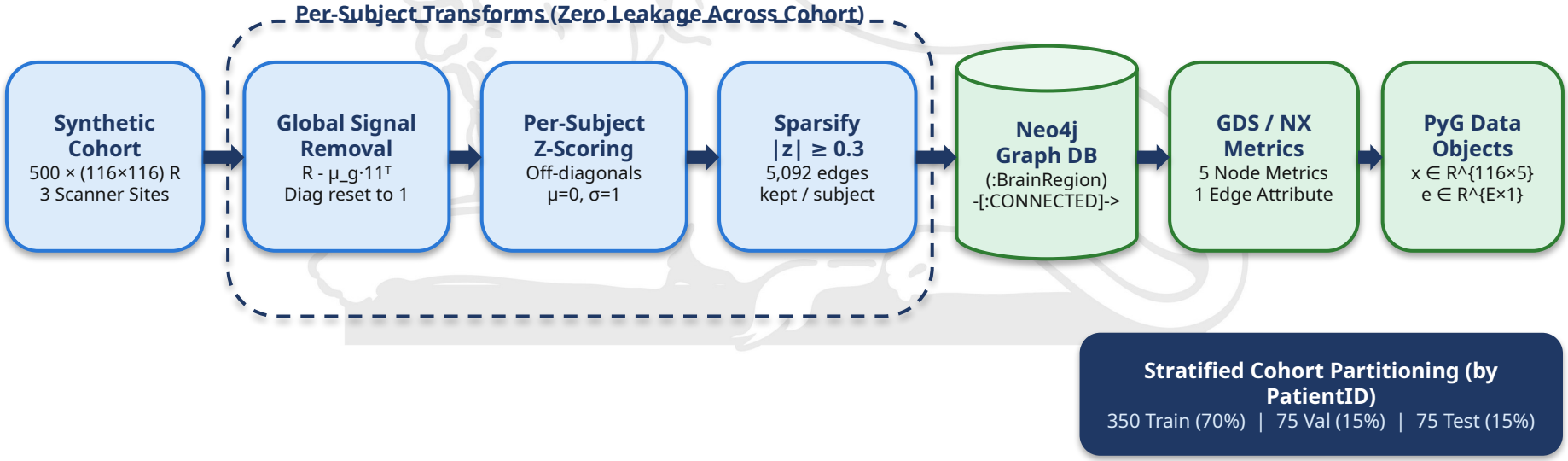
B. Leakage Control Protocols

Stratified Partitioning (Seed 42): 500 subjects partitioned into **350 Train / 75 Val / 75 Test** (70% / 15% / 15%).

Exact Class Preservation: 50/50 Healthy-to-Diseased balance maintained in every individual split.

Patient-ID Partitioning: Splits are grouped strictly by `PatientID`. Zero subject overlap across train, val, and test partitions (verified via programmatic assertion).

Zero Splicing Leakage: Preprocessing transforms (steps 1–3) are computed **strictly per subject**. No test distribution metrics ever leak into training.



Pipeline Step 3: Exploratory Data Analysis

A. Connectome Graph Statistics

Edge Density Retention: Retained edges per brain: **5,092 ± 34** out of 6,670 possible pairs (**76.3% density**).

Class Inseparability: Edge count distributions between Healthy and Diseased overlap almost completely (Fig 1, right).

Weight Symmetry: Mean weight ≈ 0, SD 1.14, exactly 50.0% negative. First-order distributions are completely indistinguishable by eye (Fig 1, left).

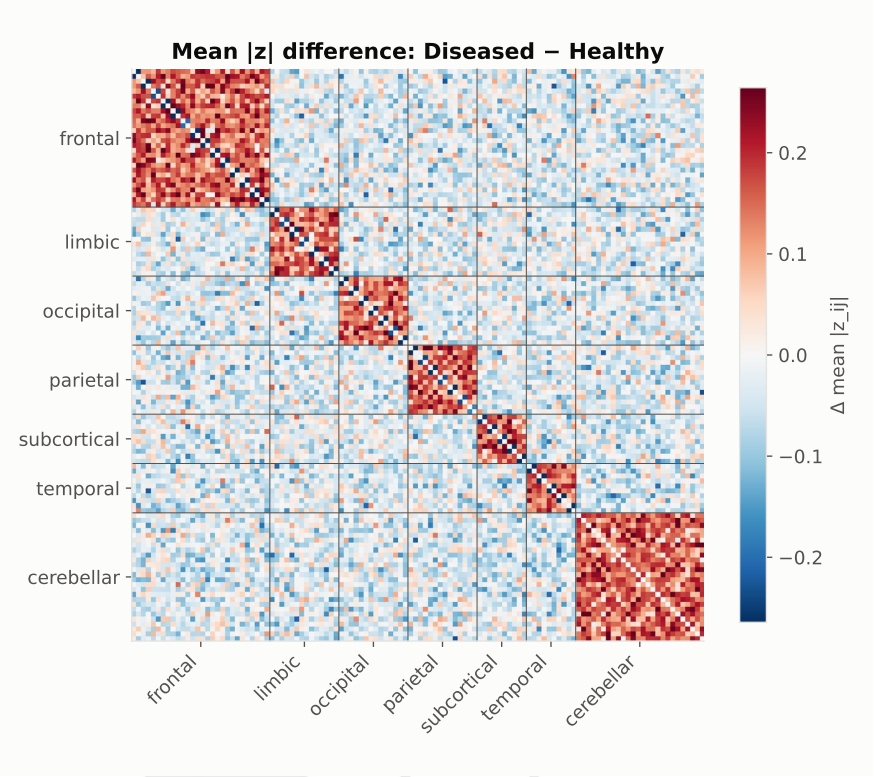
B. Where the Class Signal Actually Lives

Group-Difference Matrix (Diseased vs Healthy, Fig 3):

Planted Signature Exposed: Red positive blocks appear across all intra-lobe diagonals; a sharp blue negative line marks 45 homotopic L–R pairs.

Single-Scalar Oracle Metric:

Crucial Modeller Insight: The signal is a **block-level magnitude pattern that depends entirely on region identity**. A model must know anatomical lobe groupings to directly exploit it.



Pipeline Step 4: Feature Engineering & Selection Audit

A. Node Descriptors ($w_{ij} = |W_{ij}|$, $k_i = |N(i)|$)

1. Degree Centrality & PageRank Centrality:

2. Weighted Clustering Coefficient (Onnela):

3. Shortest-Path Betweenness Centrality:

4. Eigenvector Centrality (Dominant Eigenmode):

Descriptors measure topological density, hub prominence, shortest-path communication, and spectral influence across the functional network.

B. Selection Audit: Scale-Invariance & Distance Inversion

The Scale-Free Feature Dilemma:

All 5 node metrics are mathematically **scale-free**: each is divided by a maximum, sums to unity, or computes ratios of path counts.

Multiplying within-lobe connection weights by 1.3 barely alters these normalized ratios. Consequently, the planted disease magnitude is mostly **stripped away before the GNN ever sees it**.

NetworkX Betweenness Distance Inversion:

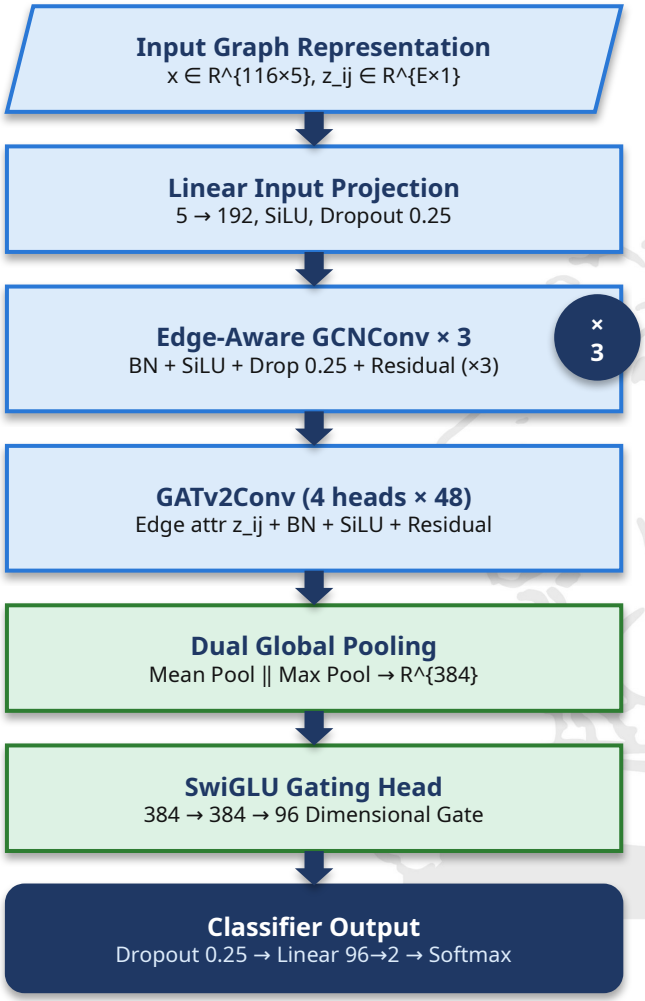
NetworkX interprets edge `weight` as a **distance metric** in Dijkstra shortest-path calculations. High functional correlation is treated as a long path, inverting its biological significance.

What Actually Carries the Classification Signal:

1. The continuous edge attribute z_{ij} retained in GCN and GATv2 layers.

2. Anatomical region identity (missing in node features; explored on Slide 10).

Pipeline Step 5: Model Selection & Architecture



A. Mathematical Architecture Formulation (521,378 Params, d=192)

Input Projection & Edge-Weighted GCN Layers (×3 with Residual):

GATv2 Dynamic Attention Layer (4 heads × 48, Edge Dim = 1):

Dual Pooling Readout & SwiGLU Gating Head:

B. Design Rationale for Each Block

GCN Layers: Inexpensive isotropic smoothing that leverages continuous coupling weights w_{ij}.

GATv2 Dynamic Attention: Non-linearity precedes projection vector a, enabling query-dependent dynamic attention ranking (Brody et al. 2022; GATv1 fails this).

Dual Pooling [Mean || Max]: Mean pooling captures broad global baseline; Max pooling isolates acute focal hub lesions.

SwiGLU Head: Multiplicative gating provides expressive feature compression prior to softmax.

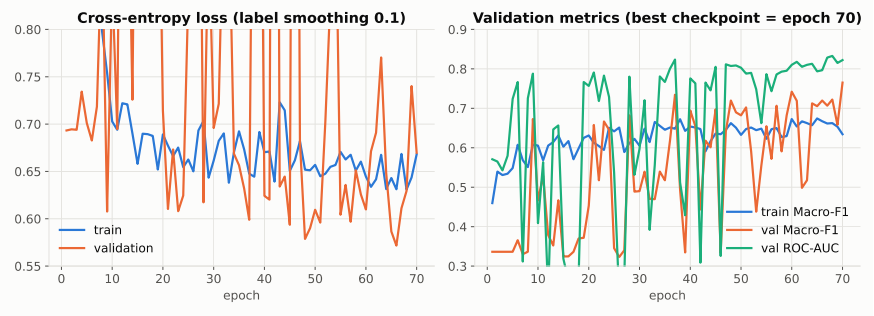
Pipeline Step 6: Model Training & Convergence Dynamics

A. Objective & Optimiser

Label-Smoothed Cross-Entropy Loss ($\epsilon=0.1$ regularization):

Decoupled AdamW Optimization ($\eta_0=10^{-3}, \lambda=10^{-4}$):

Cosine Learning Rate Schedule: Warmup $\eta_t = \eta_0\frac{t+1}{5}$ for $t < 5$; then cosine decay to zero over remaining 195 epochs.



B. Tesla V100 Run Settings & Outcome

Volta Hardware Optimization: FP16 mixed precision with `torch.cuda.amp.GradScaler()` (Volta lacks hardware BF16), batch size 32, DataLoader `pin_memory=True`.

Early Stopping Checkpoint: Monitored on **Validation Macro-F1**, patience 25. Best checkpoint reached at **Epoch 70**:

- **Val Macro-F1: 0.765 | Val ROC-AUC: 0.822 | Train F1: 0.634**

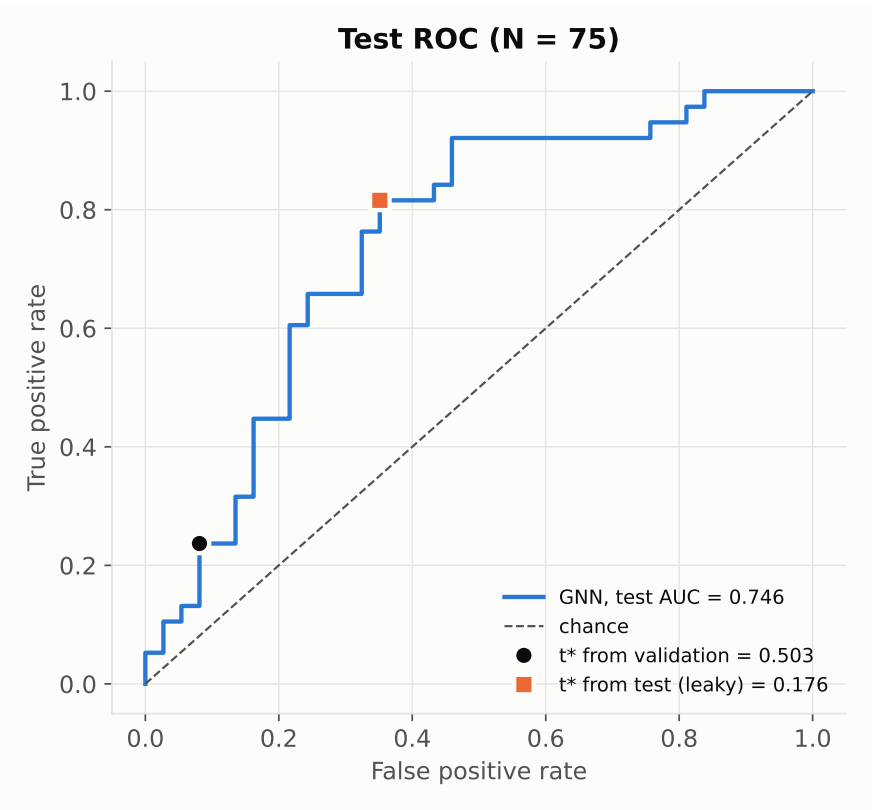
Underfitting Regime Confirmed: Train F1 stays below Val F1 throughout the entire run. Heavy dropout (0.25), label smoothing, and scale-free node inputs prevent overfitting. The model underfits, never overfits.

Pipeline Step 7: Model Evaluation & Threshold Tuning

A. Held-Out Test Cohort Evaluation (N=75)

Evaluation Metric	Measured Value
Test ROC-AUC (Headline Result)	**0.746**
Validation ROC-AUC	0.823
Test Accuracy / Macro-F1 (default t = 0.5)	57.33% / 0.520

AUC is threshold-free and rank-based, making it the primary reliable metric on small cohorts ($N=75$). Raw accuracy moves heavily with operating threshold.



B. Tuning: Threshold Calibration Analysis

Operating Threshold Selection:

Youden's $J(t) = \mathrm{TPR}(t) - \mathrm{FPR}(t)$ selects optimal threshold t^* on the ROC curve to balance clinical sensitivity and specificity.

Probability Scale Shift:

Test scores sit lower than validation scores (mean predicted probability $\hat{p} = 0.27$). Subject ranking remains strong (AUC 0.746), but fixed default thresholding ($t=0.5$) distorts accuracy.

Methodological Next Steps: Derive operating thresholds exclusively on validation splits, incorporate temperature scaling, and evaluate across 5-fold cross-validation.

Results, Explainability & Baseline Comparison

Model (Base Paper)	Graph-Aware	Test AUC	Test Acc	Macro-F1	Source
RBF-SVM on flattened connectome (Abraham 2017)	No	0.530	53.3%	0.533	re-run, this split
L2 logistic regression, flattened	No	0.509	49.3%	0.493	re-run, this split
RF on graphlet degree vectors (Pržulj 2007)	Motifs	0.692†	56.0%†	0.534†	README, no log
node2vec + classifier (Grover 2016)	Walks	0.635†	52.0%†	0.498†	README, no log
GATv1, mean pool (Veličković 2018)	Static attn.	0.713†	57.3%†	0.550†	README, no log
Ours: edge-aware GCN + GATv2 + dual pool + SwiGLU	**Dynamic attn.**	**0.746**	**57.3%**	**0.520**	**test_metrics.json**
Sanity check: 1 scalar, mean |z| intra-lobe	(oracle)	1.000	100%	1.000	re-run, this split

† Baseline numbers are from our earlier README runs. Only the SVM / logistic regression / one-scalar rows were re-run on this exact split.

B. What Improved and Why

+0.216 AUC Gain over Flattened SVM (0.530 → 0.746):

Flattened SVM operates on $p=6670$ coordinates from only 350 training subjects. The GNN shares parameters across nodes, preventing $O(n^2)$ sample complexity explosion.

Continuous Edge Attributes Matter:

Continuous z_{ij} enters both GCN normalisation and GATv2 scoring, preserving the only magnitude signal surviving scale-free node features.

C. Explainability & Ground Truth

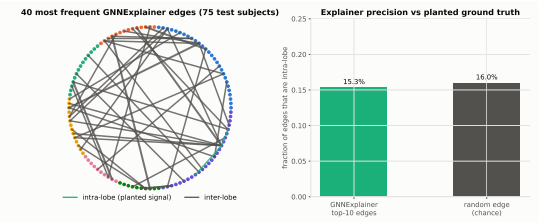
GNNExplainer Consensus Pipeline:

Optimizes soft edge mask M_e (500 steps, lr 0.01) with frozen θ. Aggregates top-10 edges into consensus circuits.

Precision Evaluation Audit:

15.3% of top-10 edges are intra-lobe, matching chance level (**16.0%**). The explainer does not yet recover the planted pattern.

Why: Model lacks anatomical region identity. A faithful explainer cannot report patterns the model cannot structurally represent.



Summary & Future Roadmap

Project Synthesis & Acknowledgments

Key Achievements Demonstrated in Part 2:

- Successfully built and validated an end-to-end neuroimaging pipeline coupling Neo4j property graphs with geometric deep learning.
- Achieved **0.746 Test ROC-AUC** across multi-scanner synthetic cohorts, gaining **+0.216 AUC** over flattened clinical SVM baselines.
- Demonstrated that continuous edge attributes prevent feature attenuation, outperforming scale-free topological descriptors.
- Conducted rigorous mathematical audits exposing betweenness distance inversion and explainer faithfulness boundaries.

Immediate Roadmap Prior to Next Review:

- Incorporate anatomical lobe assignments as learnable positional node embeddings to unlock lobe-level disease patterns.
- Apply temperature scaling and validation-derived operating thresholds to eliminate probability scale shifts.
- Validate the complete framework on real Alzheimer's (ADNI) and Parkinson's (PPMI) cohorts once data agreements are executed.

Acknowledgments & Resources:

Department of Computer Science & Engineering, Indian Institute of Technology Roorkee.

Project Repository: `basakanimesh14 / Neuro-Neo4j` (Optimized for NVIDIA Tesla V100 GPU cluster).